

Ashwin Sridharan

Platform Engineer | Kubernetes | AI Infrastructure | Distributed Systems

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PROFESSIONAL SUMMARY

Platform Engineer with **3+ years of experience** designing and building **cloud-native platforms, Kubernetes infrastructure, CI/CD systems, and self-service developer platforms** across **AWS, Azure, GCP, and on-prem environments**. Experienced in **Kubernetes platform engineering, infrastructure automation, distributed systems, observability, and reliability engineering**. Built **AI-powered engineering systems** using **Azure OpenAI, RAG, and LangGraph**, including operational tooling for **CI/CD and platform investigations**. Focused on **AI infrastructure, LLM inference, GPU orchestration, workload scheduling, and building scalable platforms** that abstract infrastructure complexity for engineering teams.

EMPLOYMENT HISTORY

Johnson Controls

05/2023 – Present | Pune

Graduate Engineer → Developer 1 → Developer 2 → Developer 3

Platform Engineering

- Architected and operated **multi-cloud Kubernetes platforms** across AWS, Azure, GCP, and on-prem environments, supporting business-critical workloads with **99.9%+ availability**.
- Engineered a **Kubernetes-native CI/CD platform** using Tekton, ArgoCD, Buildpacks, and GitOps, enabling **50+ engineers** to use standardized self-service software delivery workflows.
- Built reusable deployment workflows with **health checks, canary rollouts, automated rollback, and release orchestration**, reducing release cycles from hours to **under 15 minutes**.
- Developed Git-based change detection for monorepositories to identify impacted services and trigger **targeted pipeline execution**, reducing unnecessary builds and deployments.

AI Infrastructure & Developer Platforms

- Built an **AI-powered Platform Operations Assistant** using **LangGraph and Azure OpenAI**, enabling engineers to investigate CI/CD and platform issues through natural-language queries.
- Designed stateful agent workflows with **conditional tool routing, document retrieval, SonarQube analysis, PipelineRun lookup, and pipeline-log inspection**, enabling contextual investigation across multiple engineering systems.
- Implemented **asynchronous PostgreSQL checkpointing, thread-based session state, checkpoint recovery, bounded context windows, caching, and concurrency controls** for reliable agent execution.
- Architected an **AI-powered Security Remediation Agent** integrated with GitHub, automatically identifying vulnerabilities and generating remediation pull requests.
- Built AI-powered developer productivity solutions** using **Azure OpenAI and RAG**, enabling self-service troubleshooting, operational knowledge discovery, and engineering support.

Reliability & Security

- Led production readiness reviews for **25+ business-critical services**, validating scalability through load testing, capacity planning, and performance benchmarking.
- Owned response and resolution for **20+ Po/P1 incidents**, reducing MTTR by **40%** through deep analysis of Linux, Kubernetes, and application-level failures.
- Defined and enforced **SLOs, SLIs, and error budgets**, implementing Prometheus, Grafana, Loki, and Alertmanager and reducing alert noise by **60%**.
- Embedded security controls into CI/CD pipelines using **Trivy, SonarQube, ArmorCode, Finite State, and SBOM validation**.
- Integrated **AI-assisted vulnerability remediation workflows**, accelerating vulnerability analysis and reducing manual effort.

Cloud & Infrastructure Engineering

- Managed **multi-cloud infrastructure as code** using **Terraform and Ansible**, achieving **<5-minute RTO** while improving auditability and compliance.
- Delivered **25–30% cloud cost savings** through **resource rightsizing, lifecycle automation, and infrastructure optimization**.

Virtusa

01/2023 – 05/2023 | Chennai

Software Engineering Intern

- Developed and maintained **Java Spring Boot microservices**, implementing efficient data access layers using JDBC.

- Containerized applications using **Docker**, gaining hands-on experience with cloud-native development and deployment practices.
- Improved application responsiveness through **AJAX-based asynchronous communication** and backend performance optimizations.

AI & PLATFORM PROJECTS

GPUForge — Kubernetes-Native GPU Infrastructure Platform

Go · Kubernetes · Kueue · NVIDIA GPU Operator · vLLM · Prometheus · Helm

- Building a **Kubernetes-native GPU platform in Go** for multi-tenant **LLM inference and distributed training**, exposing self-service workloads through CRDs and APIs.
- Implementing **GPU-aware scheduling** with Kueue, including **quotas, priority, preemption, gang scheduling, and topology-aware placement**.
- Integrating the **NVIDIA GPU Operator** with **GPU sharing, MIG/MPS, NVLink/NVSwitch, and NUMA-aware resource placement** for distributed workloads.
- Building an **LLM inference platform with vLLM**, including request routing, autoscaling, canary rollouts, KV-cache-aware metrics, GPU observability, and tenant-level cost attribution

LLM-D — Disaggregated LLM Inference (Technical Research / Prototype)

LLM Inference · KV Cache · GPU Architecture · Kubernetes · GPU Networking

- Designed a **prefill/decode disaggregated inference architecture**, analyzing the different compute and memory characteristics of LLM prefill and decode workloads.
- Explored **KV-cache transfer and lifecycle management**, including GPU memory pressure, cache locality, and the trade-offs introduced by disaggregated inference.
- Analyzed GPU communication paths including **PCIe, NVLink, NVSwitch, RDMA, InfiniBand/RoCE, and GPU-aware data movement** for distributed inference.
- Evaluated inference performance across **TTFT, inter-token latency, throughput, KV-cache utilization, and GPU memory pressure**.

EDUCATION

Bachelor of Science, Data Science and Applications (Online degree)

04/2022 – 12/2025 | Chennai, India

Indian Institute of Technology, Madras 

GPA: 8.04/10

Bachelor of Technology, Information Technology

01/2019 – 05/2023 | Chennai, India

Anna University 

GPA: 9.28/10

Achievements: Department Topper · University 9th Rank

CERTIFICATIONS

- Cloud Associate Engineer (ACE), Google 
- Professional Cloud Architect (PCA), Google 
- AWS Certified Solutions Architect – Associate 

SKILLS

Kubernetes & Platform: Kubernetes, CRDs, Operators/Controllers, controller-runtime, Docker, Helm, Kustomize, ArgoCD, GitOps, Tekton, Buildpacks, RBAC

AI Infrastructure & GPU: Kueue, NVIDIA GPU Operator, vLLM, GPU Scheduling, GPU Sharing, MIG, MPS, CUDA, NCCL, KV Cache, LLM Inference, Prefill/Decode Disaggregation

Cloud & Infrastructure: AWS, Azure, GCP, Terraform, Ansible, Infrastructure as Code, Multi-Cloud

AI & LLM Systems: Azure OpenAI, LangGraph, LangChain, LangSmith, RAG, Vector Search, Agentic AI

Systems & Networking: Linux Internals, TCP/IP, DNS, CNI, Multus, SR-IOV, RDMA, InfiniBand, RoCE, Load Balancing, Distributed Systems

Reliability & Observability: SLOs, SLIs, Error Budgets, Prometheus, Grafana, Loki, Alertmanager, Capacity Planning

Programming & Data: Go, Python, Bash, SQL, Java, PostgreSQL, pgvector, Apache Kafka, Redpanda